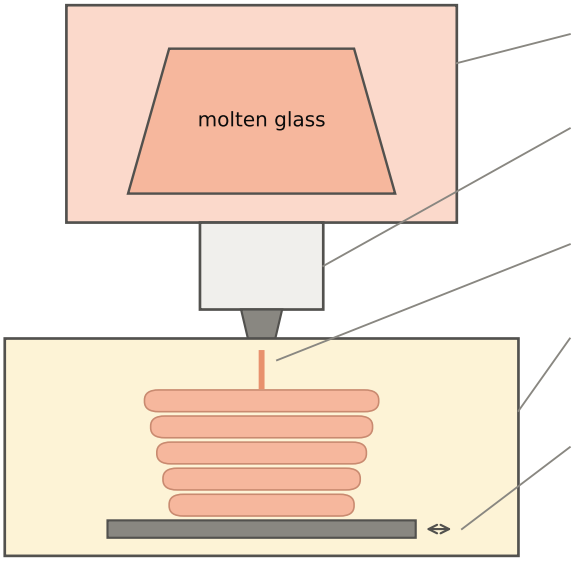

Zone 1 · melter kiln and crucible

1040–1200 °C; 2 kg (G3DP) to 25 kg (G3DP2) of soda-lime

Zone 2 · heated ceramic nozzle

800–1010 °C, bore 10–11.5 mm; flow set by melt head and temperature, started by a torch and frozen by an air ring

Thermal camera through a glass-ceramic window

7.9 µm band reads the cord surface temperature

Zone 3 · build chamber

480–525 °C, just below the annealing point, so each layer stays rigid but stress relaxes

Build plate on X–Y–Z (and rotary) axes

travel speed sets bead width; part is one continuous path

Process chain: deposition rate alone does not give production rate

Condition melt → Deposit cord → Soak in chamber → Anneal in a separate lehr → Finish and measure